POWER7

E-Commerce Project

SPECIFICATION & PROTOTYPE

P2

Presentation 2

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2026 / 2027

Tech stack & product foundation

Backend

C# / ASP.NET Core 10

01

Frontend

HTML • CSS • Bootstrap 5 • JS

02

Database

SQLite + Entity Framework Core

03

Payments

Stripe Test Mode (Sandbox)

04

Discovery & accounts - REQUIREMENTS A1–A4

A1

Catalog

Paginated catalog, max 24 items, category pages.

A2

Search

Basic search via Name attribute.

A3

Cart

Persistent cart in DB, server-validated stock.

A4

Accounts

Registration, login with Identity, address storage.

Checkout & orders - REQUIREMENTS A5–A8

A5

Checkout

Delivery address, multiple shipping methods, order total.

A6

Payment

Stripe sandbox integration via Webhook.

A7

Stock

Decrement on payment, restore on cancel, pessimistic lock.

A8

Order history

Past orders view with status and totals.

Admin, security & quality - REQUIREMENTS A9–A12

A9

Admin panel

Manage products/categories, update order status.

A10

Security

CSRF protection, server-side validation, role-based auth.

A11

Responsive UI

Mobile-first Bootstrap design down to 360px.

A12

Tests

Minimum 10 automated unit/integration tests with xUnit.

Selected bonus features

B1 • Filtered Search

B1

Combine category, price-range and sorting options directly in the URL.

B6 • Transactional Email

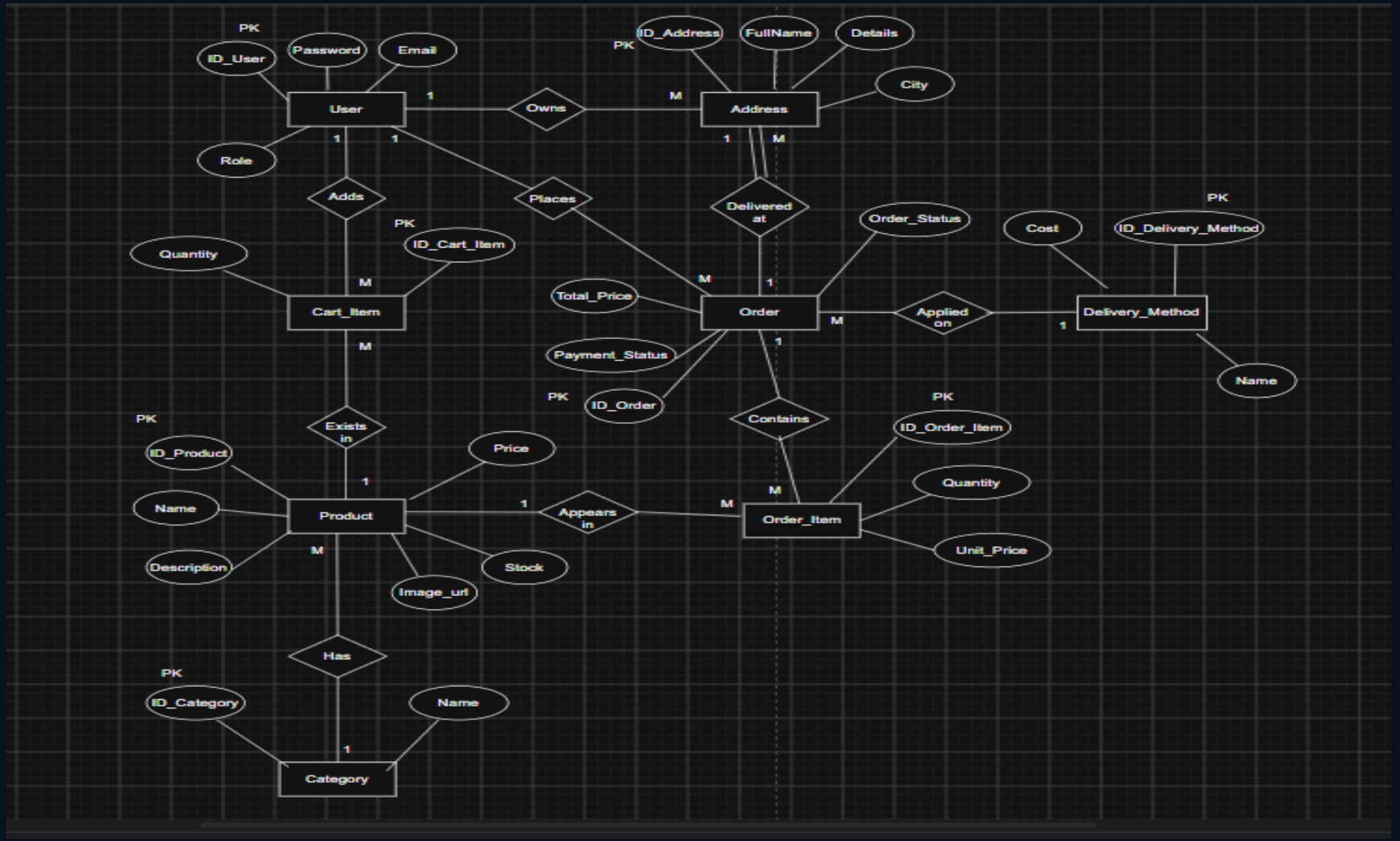
B6

Automated order-confirmation emails via Mailtrap after payment success.

Small additions, meaningful impact

- Improves product discovery and usability
- Extends the checkout workflow beyond payment
- Adds realistic e-commerce behaviour for the prototype

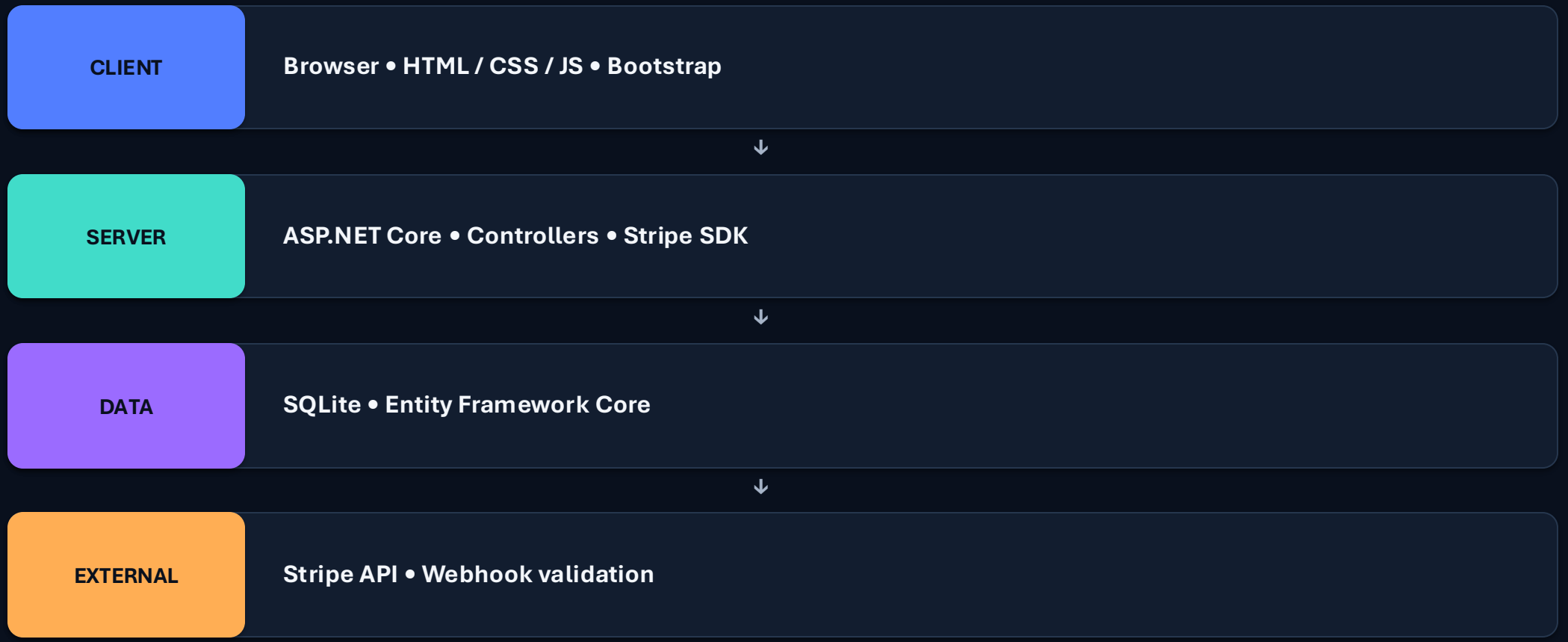
Entity-Relationship Diagram



Relational model



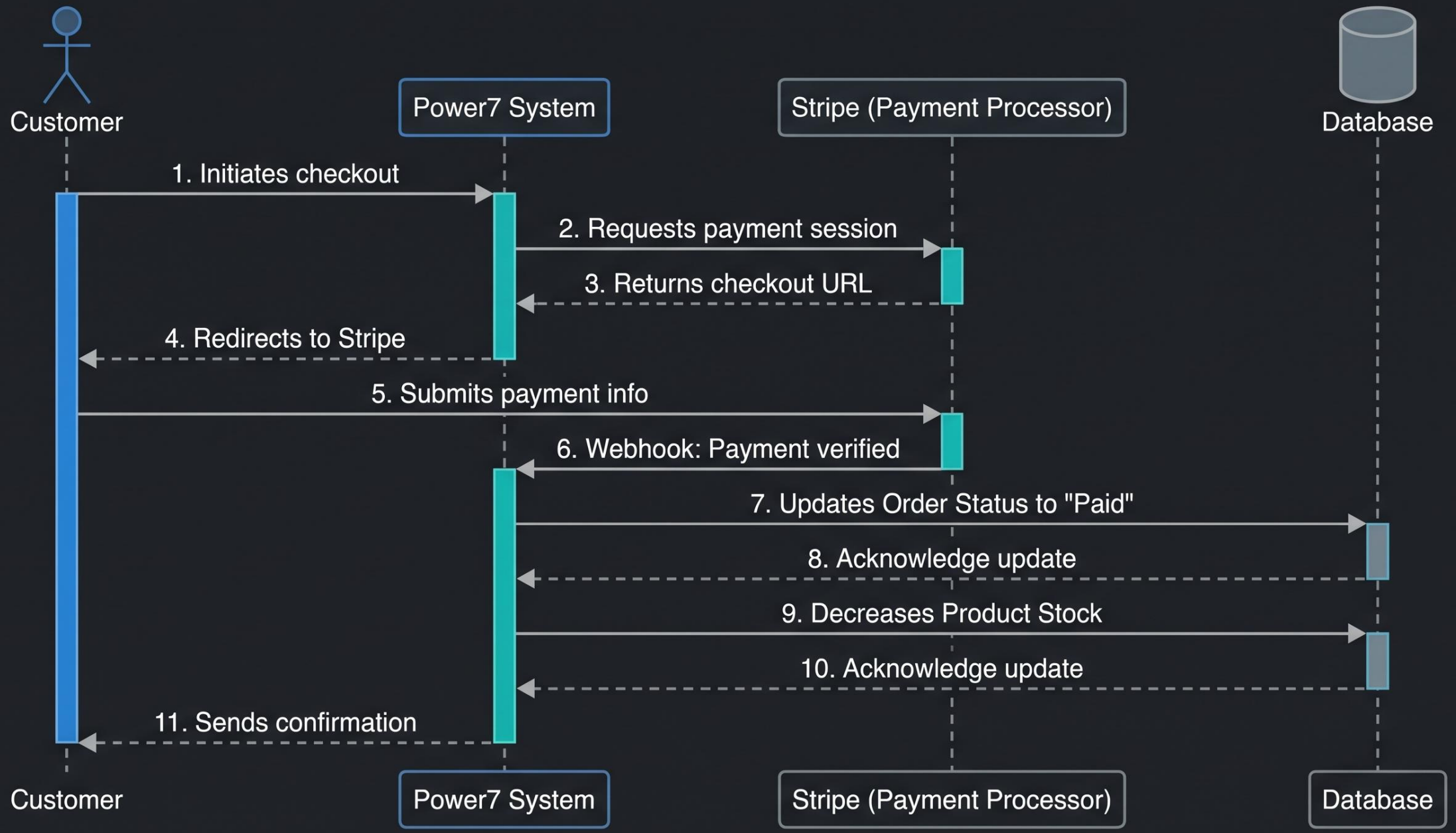
System architecture



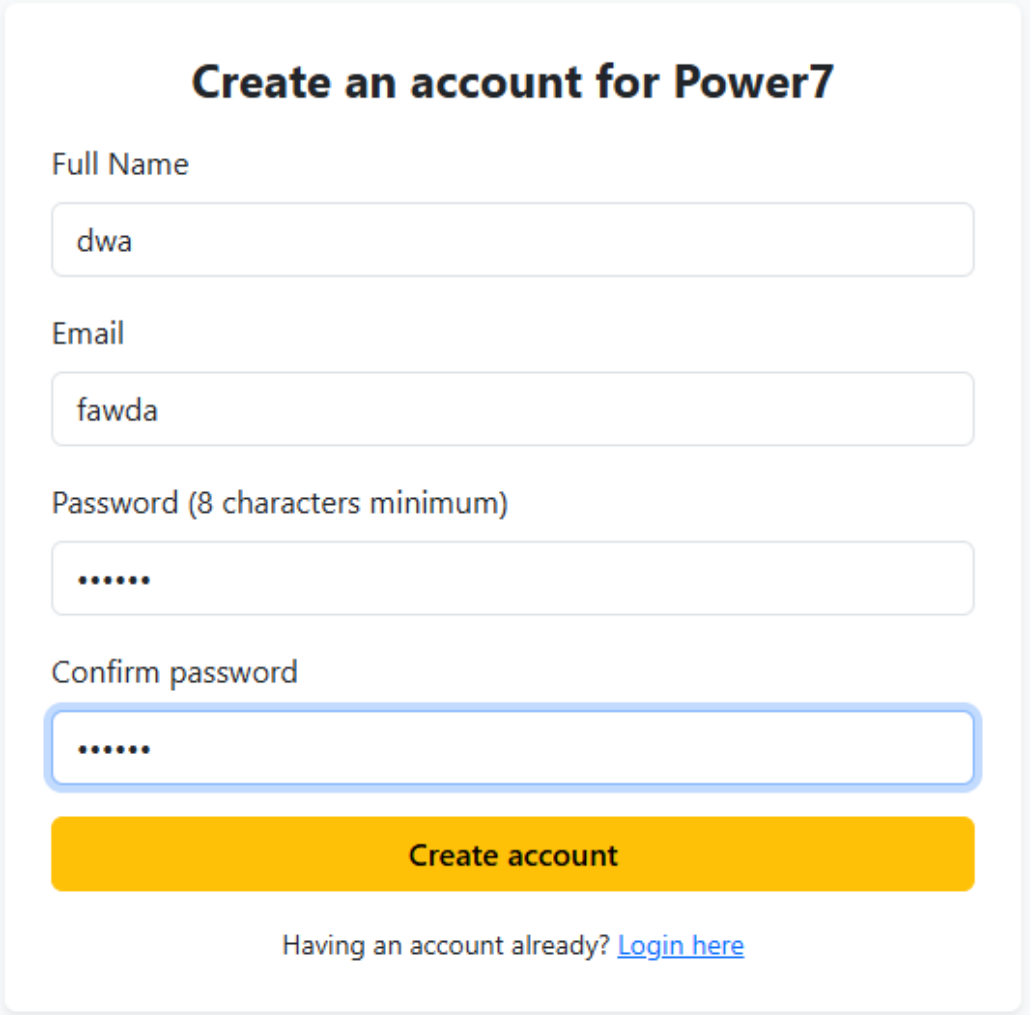
Example of endpoints

AREA	ENDPOINT	PURPOSE
PUBLIC	GET /api/products	Product catalog
AUTH	POST /api/auth/register	Create account
AUTH	POST /api/auth/login	Authenticate user
CLIENT	GET /api/cart	Persistent cart
CLIENT	POST /api/checkout/session	Create payment session
CLIENT	GET /api/orders/history	Past orders
ADMIN	/api/admin/products	Manage products
ADMIN	/api/admin/orders/{id}/status	Update order status
EXTERNAL	POST /api/webhook/stripe	Stripe webhook

Power7 System: E-Commerce Payment Flow



Wireframe example



Team ownership

Ponoran Claudiu Mihai

FRONTEND

Frontend & UI

- Catalog & product pages
- Responsive design
- Wireframes
- Bootstrap implementation

Oprea Iulian Marian

BACKEND

Backend, DB & Logic

- SQLite database & EF Core models
- Seed data
- Cart logic
- Stripe integration & API endpoints